Unit 7 Anaerobes

Introduction

Upon completion of this course, the learner will have a basic understanding of the basic concepts of cultivation and identification of *Anaerobes*

Agenda

This unit will take approximately 9 hours to complete.

This unit contains the following lessons:

Lesson	Time
Anaerobes	7 Hours
Written Test 6 (Unit 7)	2 Hours

Lesson: **7.1 Anaerobes**

Lesson Objective: **7.1.1 Differentiate between the various anaerobic bacteria**

Topics:
Concepts of Anaerobic Bacteria
Clinical Significance
Specimen Processing and Transport
Laboratory Diagnosis
Antimicrobial Susceptibility Tests
Treatment

Topic Objectives:
Describe terms associated with anaerobic bacteria to include their atmospheric requirements
Describe why anaerobic bacteria are sensitive to oxygen
Describe where anaerobic bacteria might be found in the environment to include the human body
Describe factors that predispose patients to anaerobic infections
Describe indications of anaerobe involvement in human disease
Describe gram-positive, spore-forming anaerobic bacilli to include their clinical infections
Describe gram-positive, non- spore-forming anaerobic bacilli to include their clinical infections
Describe anaerobic gram-negative bacilli to include their clinical infections
Describe anaerobic cocci to include their clinical infections
Describe acceptable specimens for anaerobic culture
Describe how anaerobic specimens are transported to include processing for recovery of anaerobic pathogens
Describe the macroscopic and microscopic examination of anaerobic cultures
Describe the media of choice for the cultivation of anaerobes
Describe the incubation conditions for cultivation of anaerobes
Describe the colony morphology and pigmentation of anaerobes
Describe the presumptive identification of anaerobes based on the results of key biochemical laboratory results
Describe definitive identification of anaerobes
Describe the identification of Clostridium species
Describe the identification of gram-positive, non- spore-forming anaerobic bacilli
Describe the identification of anaerobic gram-negative bacilli
Describe the identification of anaerobic cocci
Discuss antimicrobial susceptibility testing of anaerobes
Describe treatment of anaerobe-associated diseases

Topic: Concepts of Anaerobic Bacteria

Topic Objectives: Describe terms associated with anaerobic baceria to include their Atmospheric requirements
Describe why anaerobic bacteria are sensitive to oxygen
Describe where anaerobic bacteria might be found in the environment to include the human body
Describe factors that predispose patients to anaerobic infections
Describe indications of anerobe involvement in human disease

Introduction

- Anaerobes of clinical importance and their role in disease
    - Serious and fatal infections and intoxications
    - Endogenous bacteria
- Proper techniques for recovery and identification of anaerobes
- Procedures for susceptibility testing
- Treatment

Anaerobes Defined

- Anaerobes
    - Able to replicate in the absence of oxygen (O_2)
- Capnophilic
    - Increased (5% to 10%) carbon dioxide (CO_2), reduced (15%) oxygen (O_2)
- Microaerophilic
    - Oxygen reduced to 5% or less
- Facultative anaerobes
    - Prefers to use oxygen if available but can grow well without oxygen
- Obligate (strict) anaerobes
    - Killed in the presence of oxygen
        - Grow only in the absence of oxygen
        - Free radicals in oxyegen lead to cell destruction
            - Bactericidal
                - Antimicrobial agent that kills a bacterial isolate

- Aerotolerant (moderate anaerobes)
    - Oxygen steals electrons used in metabolism
        - Bacteriostatic
            - Antimicrobial that inhibits bacterial growth but does not kill the bacteria

- Oxygen has a direct toxic effect
    - Anaerobes lack the enzymes to break down toxic products:
        - Superoxide anion
        - Hydrogen peroxide
        - Hydroxyl radical

- Strict aerobes and facultative anaerobes can protect themselves
    - Possess enzymes to break down toxic products
        - Superoxide dismutase
        - Catalase

Where Anaerobes Are Found

- Exogenous
    - Organisms exist outside of the bodies of animals
    - Infections develop when the agent is ingested or has gained entrance through trauma
    - Example: *Clostridium, Sarcina, Fusobacterium*, and others

- Endogenous
    - Organisms exist inside the bodies of animals and become sources of infection
    - Example: *Bacteroides, Porphyromonas, Prevotella, Fusobacterium, Clostridium, Propionibacterium*, and anaerobic cocci

Endogenous Anaerobes Commonly Involved in Human Infections

- Actinomycosis:
    - *Actinomyces israelii,* other *Actinomyces* spp.

- Antibiotic-associated diarrhea; pseudomembranous colitis:
    - *Clostridium difficile*

- Bacteremia:
    - *B. fragilis group, Fusobacteria, Clostridia, Peptostreptococci*

- Brain abscess:
    - *Bacteroides* spp., *Prevotella* spp., *Porphyromonas* spp., *Fusobacterium* spp., *Clostridium* spp. (these infections are often polymicrobial)

- Infections of the female genitourinary tract:
    - *Peptostreptococci, Bacteroides* spp., *Clostridium* spp*.; Prevotella bivia, P. disiens, Actinomyces israelii* (IUD associated)

- Intraabdominal infections, liver abscess, peritonitis, perineal and perirectal infections:
    - *Bacteroides fragilis* group, other *Bacteroides* spp., *Fusobacterium* spp., *C. perfringens*, other *Clostridium* spp*., Peptostreptococci* (frequently polymicrobial)

- Myonecrosis:
    - *Clostridium perfringens, C. novyi, C. septicum* (80% to 95% of the cases)

- Oral, sinus, dental infections:
    - *Peptostreptococci*, *Porphyromonas* spp*., Fusobacterium* spp. (often polymicrobial)

- Aspiration pneumonia, pleuropulmonary infections:
    - *Porphyromonas* spp*., F. nucleatum*, *Peptostreptococci*, *B. fragilis* group, *Actinomyces* spp.

Anaerobes at Specific Anatomic Sites

- Skin
    - *Propionibacterium , Peptostreptococci*

- Upper Respiratory tract
    - *Peptostreptococci*, *Actinomyces, Propionibacterium, Fusobacterium, Prevotella, Veillonella*

- Gastrointestinal (GI) tract
    - *Bifidobacterium, Eubacterium/Eggerthella, Clostridium, Peptostreptococci, Bacteroides fragilis group, Bilophila, Fusobacterium, Porphyromonas, Prevotella, Sutterella, Veillonella*

- Genitourinary (GU) tract
    - *Peptostreptococci*, *Bifidobacterium, Fusobacterium, Lactobacillus, Mobiluncus, Prevotella, Veillonella*

- Oral cavity
    - *Actinomyces, Eubacterium/Eggerthella*, *Peptostreptococci*, *Fusobacterium, Prevotella, Bifidobacterium, Porphyromonas, Veillonella*

Factors That Predispose Patients to Anaerobic Infections

- Trauma to protective barriers

- Vascular stasis
    - Prevention of oxygen flow to a particular site
        - Example:
            - Inflammation of a wound, or diabetes

- Gains entry and has ideal conditions that allows for a perfect growth enviroment
    - Bite wounds
    - Aspiration of oral flora into deep lung
    - Oral surgery
    - Puncture wounds or burns with soil contamination

Table 7.1 – Potential Virulence Factors of Anaerobic Bacteria

Potential Virulence Factor	Possible Role	Anaerobes Known or Thought to Possess
Polysaccharide capsules	Promotes abscess formation; antiphagocytic function	*Bacteroides fragilis, Porphyromonas gingivalis*
Adherence factors	Fimbriae, fibrils enable organisms to adhere to cell surfaces	*B. fragilis, P. gingivalis*
Clostridial toxins, exoenzymes		
Collagenases	Catalyze the degradation of collagen	Certain *Clostridium* spp.
Cytotoxins	Toxic to specific types of cells	*C. difficile*
DNases	Destroy DNA	Certain *Clostridium* spp.
Enterotoxins	Toxic to cells of the intestinal mucosa	*C. difficile*
Hemolysins	Lyse red blood cells liberating hemoglobin	Certain *Clostridium* spp.
Hyaluronidase	Catalyzes the hydrolysis of hyaluronic acid, the cement substance of tissues	Certain *Clostridium* spp.
Lipases	Catalyze the hydrolysis of ester linkages between fatty acids and glycerol of triglycerides and phospholipids	Certain *Clostridium* spp.
Neurotoxins (e.g., botulinum toxin, tetanospasmin)	Destroy or disrupt nerve tissue	*C. botulinum, C. tetani*
Phospholipases	Catalyze the splitting of host phospholipids (lecithinase)	Certain *Clostridium* spp.
Proteases	Split host proteins by hydrolysis of peptide bonds	Certain *Clostridium* spp.

Indications of Involvement of Anaerobes in Infectious Processes

- Infection in close proximity to a mucosal surface

- Presence of foul odor

- Presence of large quantity of gas

- Presence of black color or brick red fluorescence when exposed to long-wave UV light such as a Wood's light

- Presence of sulfur granules: dense clumps of bacteria that may be colored

Topic: Clinical Significance

Topic Objectives: Describe gram-positive, spore-forming anaerobic bacilli to include their clinical infections
Describe gram-positive, non-spore-forming anaerobic bacilli to include their clinical infections
Describe anaerobic gram-negative bacilli to include their clinical infections
Describe anaerobic cocci to include their clinical infections

Frequently Encountered Anaerobes

- *Clostridium* spp. Gram-positive
- Peptococcaceae Gram-positive
- *Bacteroides* spp. Gram-negative
- *Fusobacterium* spp. Gram-negative
- *Prevotella* spp. And *Porphyromonas* spp. Gram-negative
- *Veillonella* spp. Gram-negative

Gram-positive, Spore-Forming Anaerobic Bacilli

- Genus: *Clostridium* (Clostridia)
  - *C. perfringens*
    - Boxcar, large, square rods, gram-positive bacillus
    - Produces a double zone of hemolysis
    - Subterminal spores
  - *C. tetani*
    - Slow growing
    - Heavily swarming
    - Terminal spores
  - *C. septicum*
    - Heavily swarming; swarming; Gray; circular, glossy, translucent resembling a “Medusa head”; beta-hemolytic
    - Subterminal spores
  - *C. botulinum*
    - Gray-white; circular to irregular; usually beta-hemolytic
    - Subterminal spores
  - *C. histolyticum*
    - Bread crumb
    - Subterminal spores

- *C. novyi*
  - Subterminal spores
- *C. bifermentans*
  - Subterminal spores
- *C. difficile*
  - Large, flat colonies; barnyard odor, chartreuse fluorescence (yellow-green)
  - Subterminal spores
- *C. tertium*
  - Convex, smooth
  - Aerotolerant growth
  - Terminal spores

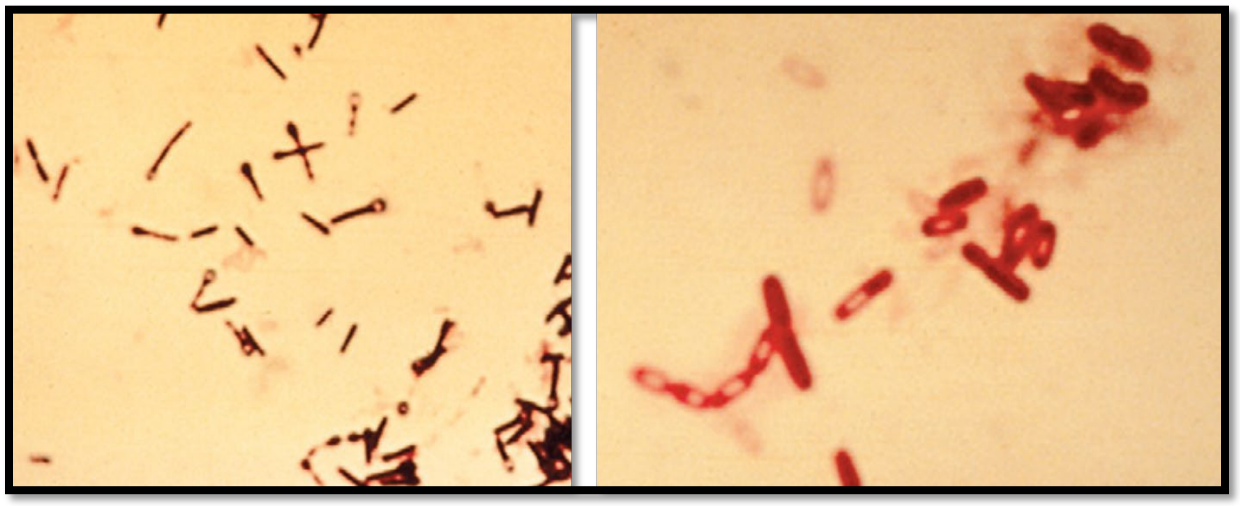


Figure 7.1 – Classification of Clostridia by Endospore Location

Clinical Infections

- Food poisoning
- Botulism
- Tetanus
- Myonecrosis
- Bacteremia
- Clostridium difficile-Associated Diseases

Food Poisoning

- Causative agent
  - *Clostridium perfringens*

- Toxins produced
  - Type A produce α toxin
  - Type C produce α and β toxins

- Symptoms
  - Type A: Diarrhea and cramps symptoms 8-30 hours after ingestion linked to sporulation
  - Type C Acute onset of severe abdominal pain and diarrhea, which is often bloody, and may be accompanied by vomiting symptoms as soon as 5- 6 hours

- Source
  - Ingestion of enterotoxin-producing strains in contaminated food

- Treatment/prevention
  - Type A: Usually self-resolving
  - Type C: Oral rehydration or, in severe cases, intravenous fluids and electrolyte replacement can be used to prevent or treat dehydration
  - Proper food handling

Botulism

- Causative agent
  - *Clostridium botulinum*

- Toxins produced
  - Botulinum toxins
  - A through G, but only types A, B, and E are associated with human disease
    - Toxin A is now used in medical treatments, such as Botox®
    - Prevents the release of acetylcholine
      - Causes paralysis and sometimes death

- Symptoms
  - Flaccid paralysis
  - Double vision, blurred vision, droopy eyelids, slurred speech, difficulty swallowing, dry mouth, and muscle weakness
  - 2 hours to 8 days later

- Source
    - Home canning
    - Home-cured meats
    - Honey
        - Infant botulism
            - Ingestion of spores in contaminated honey
            - Once ingested, the spores germinate, and the vegetative cells then colonize the colon and subsequently produce toxins

- Treatment/prevention
    - Proper food handling during manufacture
    - Antitoxin

Tetanus

- Causative agent
    - *Clostridium tetani*

- Toxins produced
    - Tetanospasmin neurotoxin
        - Acts on inhibitory neurons, preventing the release of neurotransmitters
            - Causes contortion of muscles

- Symptoms
    - Muscular rigidity of the jaw, neck, and lumbar region
        - Difficulty in swallowing
        - Sometimes involves limbs as well
    - Begins about 7 days after inoculation

- Source
    - Wounds contaminated with dirt, feces, or spit (saliva)
    - Wounds caused by an object puncturing the skin
    - Burns
    - Crush injuries

- Treatment/prevention
    - Diphtheria, pertussis, tetanus (DPT) vaccine

Myonecrosis

- Also known as *gas gangrene*
    - Infected wounds from trauma or surgery

- Causative agent
  - *C. perfringens*: most common cause
  - *C. histolyticum*
  - *C. septicum*
  - *C. novyi*
  - *C. bifermentans*

- Toxins produced
  - Alpha toxin causes tissue necrosis and deeper invasion

- Symptoms
  - Pain and swelling with obvious tissue necrosis

- Source
  - Infected wounds causes death of tissue that can require amputation

- Treatment/prevention
  - Antibiotic and surgical debridement of necrotic tissue
  - Hyperbaric oxygen can help destroy anaerobes

Bacteremia

- Causative agent
  - *C. perfringens*
    - Most common isolate from blood cultures
  - *C. septicum*
    - Marker organism for malignancy of GI tract
  - *C. bifermentans*
  - *C. tertium*

- Toxins produced
  - Alpha toxin causes tissue necrosis and deeper invasion
    - Toxin prevents the cell from prosessing essential proteins for life causing the cell to die

- Symptoms
  - Fever, a rapid heart rate, shaking chills, low blood pressure, gastrointestinal symptoms (such as abdominal pain, nausea, vomiting, and diarrhea), rapid breathing

- Source
  - Dental or medical procedures
  - Injection of recreational drugs

- Treatment/prevention
    - Antibiotics
    - Proper sterile practices

***Clostridium difficile* - Associated Diseases**

- Causative agent
    - Clostridium difficile
        - A common cause of health care–associated (nosocomial) infection
        - Most common isolate in antibiotic-associated diarrhea

- Symptoms
    - Pseudomembranous colitis
        - Necrosis of colon tissue and bloody diarrhea

- Toxins produced
    - Toxin A: enterotoxin
    - Toxin B: cytotoxin

- Sources of infection
    - Transferred among patients
    - Hands of hospital personnel
    - Virulent strain B1/NAP1/027
        - Produces larger amounts of toxins A and B, plus a third toxin, binary toxin, and has high level resistance to fluoroquinolones

- Treatment/prevention
    - Antibiotics such as metronidazole, vancomycin, or fidaxomicin
    - Hand washing and proper hygiene

Gram-positive Non–Spore-Forming Anaerobic Bacilli

- Phyla: Actinobacteria
    - *Actinomyces* spp.
        - Small, smooth, flat, convex, gray-white, translucent
    - *Bifidobacterium*
        - Small, white, convex, shiny, with irregular edge
    - *Propionibacterium* spp
        - Small gray-white and become larger and more yellowish tan with age
    - *Mobiluncus* spp.
        - Small, low, convex, translucent
    - *Eggerthella*
        - Small, gray, translucent, circular, convex
        - Catalase variable

- Phyla: Firmicutes
    - *Lactobacillus* spp.
        - Varies greatly, some species pinpoint α-hemolytic colonies on SBA
        - Others are gray, medium, rough appearance
        - Spiral-shaped
        - Usually considered a contanminant
        - Serious infections, primarily bacteremia and endocarditis, are known to occur in immunocompromised patients

- Clinical Infections
    - Actinomycosis
    - Bacterial vaginosis

Actinomycosis

- Chronic, granulomatous disease characterized by sinus tracts and fistulae that erupt to the surface and drain pus that contains sulfur granules dense clumps of bacteria that may be colored

- Causative Actinobacteria agents:
    - *Actinomyces israelii*
        - Have a molar tooth appearance. Depending on the species, colonies may be red, pink, tan, yellow, white, or grayish
    - *Bifidobacterium*
        - Variable in shape, ranging from coccobacilli to long branching rods. The ends of the cells may be pointed, bent, club-shaped, spatulated, or bifurcated (forked)
    - *Propionibacterium*
        - Variable in shape, ranging from coccoid and short diphtheroidal rods to long, branched filaments

- Symptoms
    - Abscesses, tissues fibrosis, and draining sinuses that discharge sulfur granules

- Source
    - Often follows dental sepsis or manipulation, trauma, tonsillitis, otitis, or mastoiditis

- Treatment/prevention
    - Penicillin is the preferred treatment

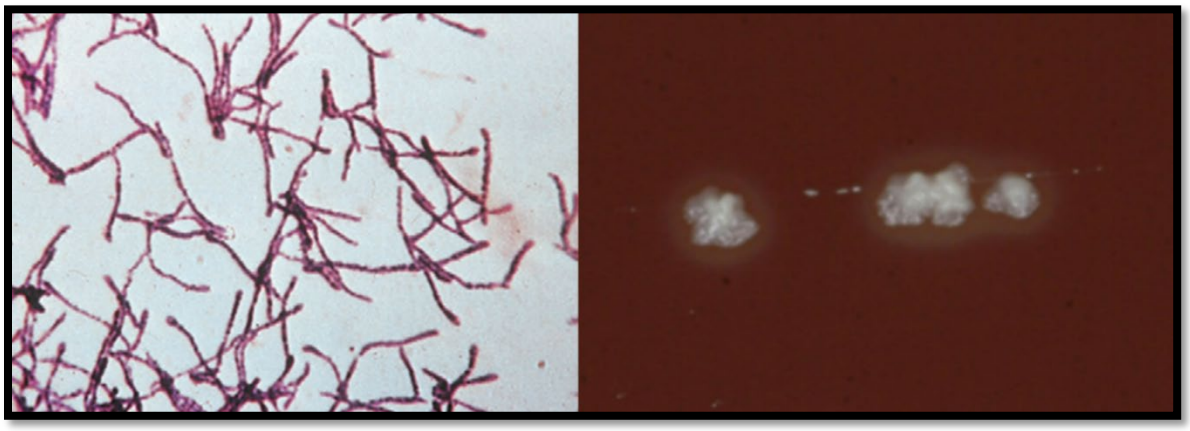


Figure 7.2 – Actinomyces israelii gram stain

Bacterial vaginosis

- *Lactobacillus* spp. Usually comprise the largest portion of the vaginal biota

- In BV, a shift in the vaginal biota occurs, resulting in the overgrowth of other endogenous anaerobes

- Causative agents:
  - *Mobiluncus* spp.
  - *Bacteroides* spp.
  - *Prevotella* spp.
  - *Gardnerella* spp.

- Symptoms
  - Gray-white, homogenous, malodorous vaginal discharge

- Source
  - Vaginal

- Treatment/prevention
  - Antibiotics

Anaerobic Gram-negative Bacilli

- Frequently associated with abscesses, peritoneal infections, diabetic foot ulcer, decubitus pressure sores

- The genera most commonly encountered in clinical specimens
    - *Bacteroides fragilis* group
        - White to gray, circular, convex, translucent to semiopaque
    - *Porphyromonas*
        - Dark brown to black, mucoid, White, circular, convex, translucent to opaque
    - *Prevotella*
        - White, circular, convex, translucent to opaque
    - *Fusobacterium*
        - Chartreuse fluorescence

- Clinical Infections
    - Bacteremia
    - Lemierre's disease

Bacteremia

- Causative agents:
    - *B. fragilis* group
        - *B. fragilis* is the most common species of anaerobic bacteria isolated from infectious processes of soft tissue and anaerobic bacteremia and is responsible for more than 60% of the infections caused by the *B. fragilis* group
    - Fusobacteria

- Symptoms
    - Fever, rapid heart rate, shaking chills, low blood pressure
    - Sepsis, severe sepsis, or septic shock
        - Abdominal pain, nausea, vomiting, and diarrhea, rapid breathing, and confusion

- Source
    - Abscesses occurring beneath mucosal surfaces, peritoneal infections, diabetic foot ulcers, decubitus pressure

- Treatment/prevention
    - *Fusobacteria*
        - Kanamycin sensitive
        - Colistin sensitive
    - *B. fragilis* group
        - Antibiotics

Lemierre's disease

- Causative agents
  - *Fusobacterium necrophorum*

- Symptoms
  - A peritonsillar abscess containing a number of bacterial species

- Source
  - A rare condition that can occur following group A streptococcal pharyngitis which leads to thrombophlebitis of the jugular vein

- Treatment/prevention
  - Kanamycin sensitive
  - Colistin sensitive

Anaerobic Cocci

- Gram-positive
  - Formerly *Peptostreptococcus*
  - Has now been reclassified into at least five different genera
    - *Peptostreptococcus*
      - Medium white to gray, opaque, sweet, fetid odor
      - *Peptococcus niger* still not settled into a genera
        - Tiny, black, convex, shiny, smooth, circular, entire edge; becoming light gray when exposed to air
    - *Anaerococcus*
      - Small, white, translucent, smooth
    - *Finegoldia*
      - Tiny, gray, translucent, nonhemolytic
    - *Parvimonas*
    - *Peptoniphilus*

- Gram-negative
  - *Veillonella* spp.
    - Small, almost translucent, grayish white, smooth, opaque

Clinical Infections

- The anaerobic cocci are weakly pathogenic but can be isolated from a wide variety of infections
  - Brain abscess
  - Meningitis
  - Aspiration pneumonia
  - Lung abscess
  - Gingivitis and other periodontal diseases

- Source
  - Most often associated with polymicrobial infections
  - Blood cultures
  - Infections following orthopedic surgery

- Treatment
  - *Peptostreptococcus*
    - Vancomycin sensitive
  - *Finegoldia*
    - Kanamycin sensitive
    - Vancomycin sensitive
  - *Veillonella* spp.
    - Kanamycin sensitive
    - Colistin sensitive

Topic: Specimen Processing and Transport

Topic Objectives: **Describe acceptable specimens for anaerobic culture**
Describe how anaerobic specimens are transported to include processing for recovery of anaerobic pathogens

Table 7.2 – Acceptable Specimens for Anaerobic Bacteriology

Source	Specimens Recommended for Collection
• Central nervous system:	• Cerebrospinal fluid, aspirated abscess material, tissue from biopsy or autopsy
• Dental, ear-nose-throat specimens:	• Aspirated abscess material, biopsied tissue
• Localized abscesses:	• Needle and syringe aspiration of closed abscesses
• Decubitus ulcers:	• Aspirated pus
• Sinus tracts or draining wounds:	• Aspirated material

Source	Specimens Recommended for Collection
• Deep tissue or bone:	• Specimens obtained during surgery from depths of wound or underlying bone lesion
• Pulmonary:	• Aspirate obtained by direct lung puncture; pleural fluid obtained by thoracentesis; open lung biopsy; sulfur granules from draining fistula
• Intra abdominal:	• Aspirate from abscess, ascites fluid, peritoneal fluid, tissue
• Urinary tract:	• Suprapubic bladder aspiration
• Female genital tract:	• Aspirate from loculated abscess; culdocentesis specimen
• Other:	• Blood, bone marrow, synovial fluid, biopsied tissue from any normally sterile site

Table 7.3 – Unacceptable Specimens for Anaerobic Bacteriology

Source	Specimens
• Dental, ear-nose-throat specimens:	• Throat swabs, nasopharyngeal swabs, Gingival swabs or any other intraoral surface swabs,
• Pulmonary:	• Sputum obtained by nasotracheal or endotracheal suction; bronchial washings or other specimens obtained via a bronchoscope (unless a protected double-lumen catheter is used); expectorated sputum
• Female genital tract:	• Vaginal, cervical, or urethral swabs; female genital tract specimens collected via the vagina; swabs of a vaginal discharge
• Urinary tract:	• Voided or catheterized urine

Source	Specimens
• Feces:	• Large bowel contents; feces; ileostomy and colostomy effluents; rectal swabs; gastric and small bowel contents
• Ulcers/ wound:	• Surface swabs from decubitus ulcers, perirectal abscesses, foot ulcers, exposed wounds, eschars, pilonidal sinuses, and other sinus tracts
• Other:	• Any material adjacent to a mucous membrane that has not been adequately decontaminated

Specimen Quality, Transport, and Processing

- Appropriate sample collection for anaerobes
    - Collect accepteble sample from site of infection
        - Prevents misleading and nonuseful clinical results

- Requires rapid processing to prevent to prevent shocking the anerobes
    - Toxicity of oxygen
    - Drying of specimen

- Aspirates
    - Collecting with needle and syringe prevents contamination and can expel any air
    - Best to inject specimen into prereduced anaerobically sterilized (PRAS) media
    - Vortex to mix sample and plate to appropriate isolation media
        - Use a few drops on PRAS media and streak for isolated colonies
        - Vital since most anaerobic infections are polymicrobial
            - Requires isolated colonies to verify relevant organisms

- Swabs
    - Not recommended and should only be collected under prereduced conditions
    - Eswab transport system is the only liquid-based multipurpose collection and transport system that maintains viability of aerobic, anaerobic and fastidious bacteria for up to 48 hours at room and refrigerated temperature

- Tissue - Place small piece into PRAS media or reduced transport medium

- Blood
    - Aseptic inoculation of both anaerobic and aerobic blood culture bottles with meticulous preparation of venipuncture site to avoid contamination



Figure 7.3 – Specimen Quality, Transport, and Processing

Topic: Laboratory Diagnosis

Topic Objectives: **Describe the macroscopic and microscopic examination of anaerobic cultures**
Describe he media of choice for the cultivation of anaerobes
Describe the incubation conditions for cultivation of anaerobes
Describe the colony morphology and pigmentation of anerobes
Describe the presumptive identification of anaerobes based on the results of key biochemical laboratory results
Describe definitive identification of anaerobes
Describe the identification of *Clostridium* species
Describe the identification of gram-positive, non-spore-forming anaerobic bacilli
Describe the identification of anaerobic gram-negative bacilli
Describe the identification of anaerobic cocci

Processing Clinical Samples for Recovery

- Macroscopic and microscopic examination of specimen using gram stain
- Inoculation of appropriate plated and tubed media
- Anaerobic incubation of inoculated media

Characteristics to Note During the Macroscopic Examination of a Specimen

- Is it an appropriate specimen?
  - Inappropriate specimens should be rejected
- Was it submitted in an appropriate transport container or medium?
  - Improperly transported specimens should be rejected
- Does the specimen have a foul odor?
  - Many anaerobes, especially *Fusobacterium* and *Porphyromonas*, have foul-smelling metabolic end products
- Does the specimen fluoresce brick red when exposed to long-wave (366-nm) ultraviolet light?
  - Pigmented species of *Porphyromonas* and *Prevotella* produce substances that fluoresce under long-wave UV light prior to becoming darkly pigmented
    - Although a brick red fluorescence is presumptive evidence of these organisms, some members of this group fluoresce colors other than brick red

- Is the necrotic tissue or exudate black?
    - Such discoloration may be caused by the pigment produced by pigmented species of *Porphyromonas* and *Prevotella*

- Does the specimen contain sulfur granules?
    - Such granules are associated with actinomycosis, a condition caused by *Actinomyces*, *Propionibacterium propionicum*, and closely related organisms, such as *Propionibacterium acnes*

Direct Microscopic Examination of Specimen

- Gram stain
    - Clinical and Laboratory Standards Institute (CLSI) guidelines recommends gram stains on all anaerobic specimens
    - Establishes a polymicrobial or monomicrobial infection
    - Starts the presumptive identification process
    - Aids in selection of culture media
    - Determines the quality of specimen collected
    - Use methanol as fixative instead of heat
    - Absence of leukocytes may indicate necrotizing toxins
    - Recognize morphology with presumptive identification
        - Gram-positive bacilli
            - *Clostridium*
        - Thin gram-negative bacilli
            - *Fusobacteria*
        - Tiny gram-negative cocci that are gram variable
            - *Veillonella*
        - Gram-negative coccobacilli
            - *Bacteroides, Porphyromonas, Prevotella*

Media and media inoculation

- Primary plating media for anaerobic isolation

- Inoculation procedures

- Anaerobic incubation of inoculated media

Primary Setup Media Recommended for Recovery of Anaerobes

- Anaerobic blood agar (CDC)
    - Supports growth of almost all obligate and facultative anaerobes, best for anaerobic gram-positive cocci
    - An enriched medium containing sheep blood for enrichment and detection of hemolysis, vitamin K (required by some *Porphyromonas* spp.), and yeast extract

- Bacteroides bile esculin agar (BBE)
    - Supports growth of bile-tolerant *Bacteroides* spp., some strains of *Fusobacterium mortiferum*, *Klebsiella pneumoniae*, enterococci, and yeast may grow to a limited extent

- Brucella blood agar (BRU/BA)
    - Supports growth of almost all obligate and facultative anaerobes, best for gram-negative bacteria
    - An enriched medium containing sheep red blood cells for enrichment and detection of hemolysis, casein peptones, dextrose, yeast extract, vitamin K, and hemin

- Kanamycin–vancomycin–laked blood agar (KVLB)
    - Supports growth of Bacteroides and Prevotella spp.; yeasts and kanamycin-resistant, facultative, gram-negative bacilli will also grow

- Phenylethyl alcohol agar (PEA)
    - Supports growth of almost all obligate anaerobes (gram-positive and gram-negative) and gram-positive, facultative anaerobes

- Colistin nalidixic acid blood agar plate (CNA)
    - Supports growth of almost all obligate anaerobes (gram-positive and gram-negative) and gram-positive, facultative anaerobes

- Anaerobic broth (e.g., thioglycollate [THIO] and chopped or cooked meat)
    - Supports growth of almost all types of bacteria; in THIO, obligate aerobes and microaerophiles grow near the top, obligate anaerobes at the bottom, and facultative anaerobes throughout the broth

- Cycloserine–cefoxitin–fructose agar (CCFA) plate
    - CCFA is a selective and differential medium for the recovery and presumptive identification of *C. Difficile*
    - *C. Difficile* produces yellow ground-glass colonies
    - *C. Difficile* has a characteristic odor, resembling a horse stable

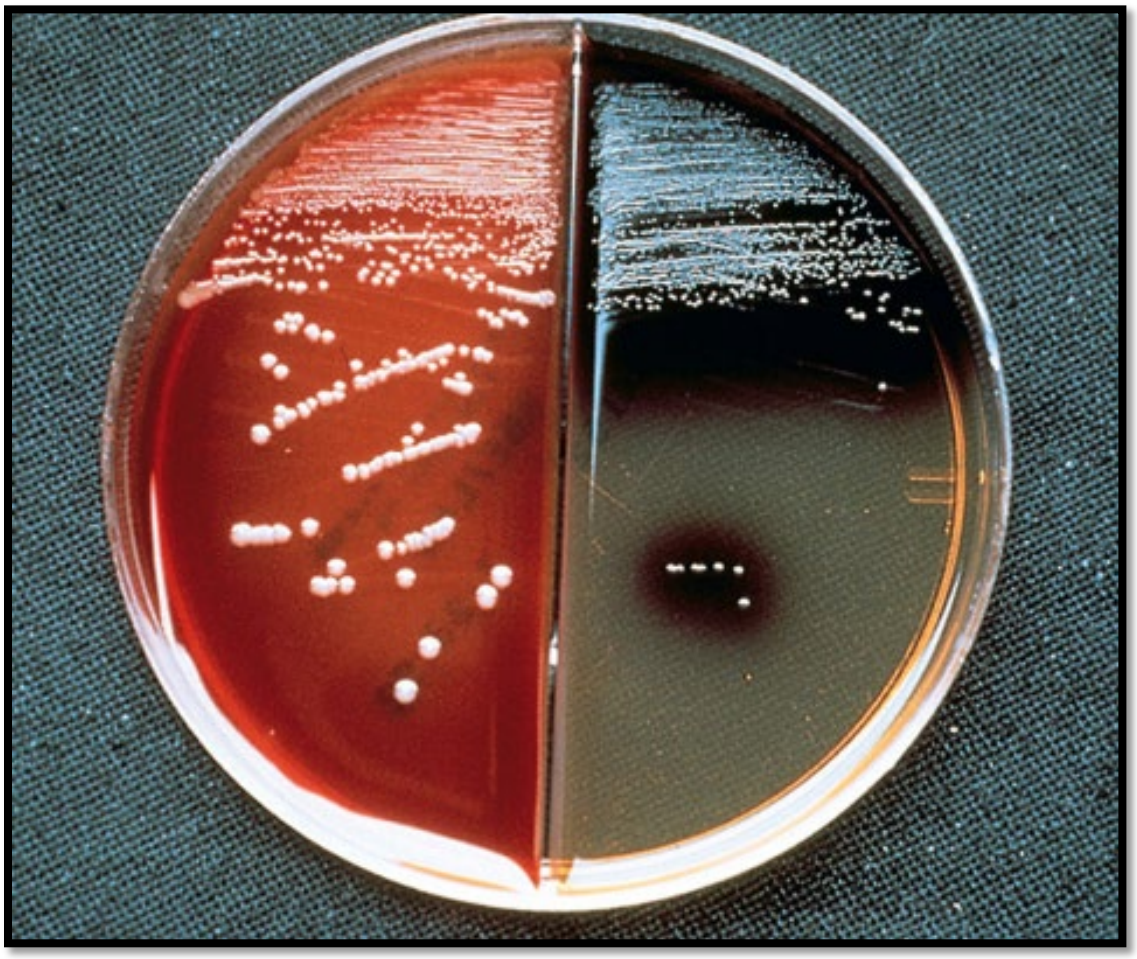


Figure 7.4 – Colonies on Kanamycin–vancomycin–laked blood agar (KVLB)

- PRAS media
    - Prepared by boiling, autoclaving, and replacing any air with an O_2 free gas mixture



Figure 7.5 – PRAS Media

Additional Media for Aerobic Incubation

- In most cases, a request for anaerobic culture on a specimen is also accompanied by a request for aerobic culture

- Plated media to be incubated aerobically (in a CO_2 incubator) is also inoculated
  - 5% sheep blood
  - Macconkey
  - Chocolate agar plates
  - CNA blood agar plate selective for gram-positive bacteria

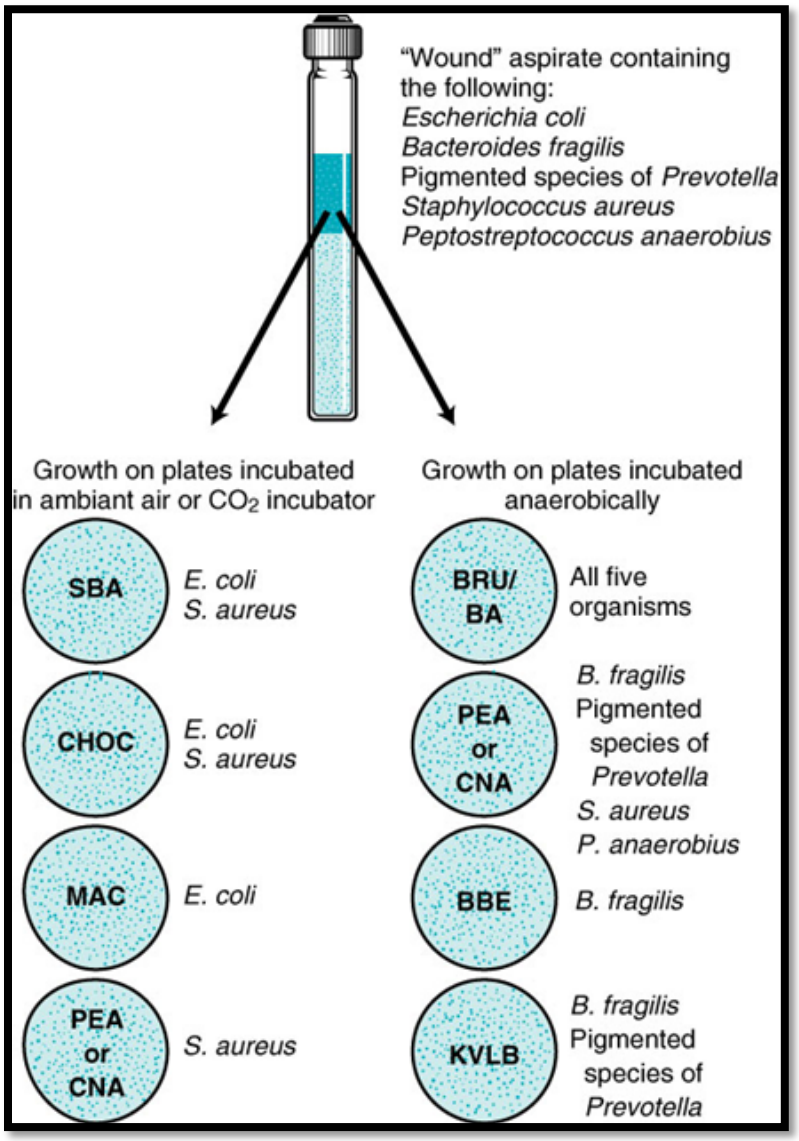


Figure 7.6 – Inoculating of aerobic and anaerobic organisms

Inoculation Procedures

- Inoculation of appropriate plated and tubed media could be performed in an area with a suitable nitrogen gas holding system or near-anaerobic conditions
    - Batch processing
        - Placing into holding system
            - A jar, box, or other small chamber

Anaerobic Incubation

- Anaerobic chambers (most ideal for incubating anaerobic media)
    - Glove boxes

- Anaerobe jars

- Anaerobic bags

Anaerobic Chambers

- Gases
    - Hydrogen gas (5%-10%)
    - CO_2 gas (5%-10%)
    - Nitrogen gas (80%-90%)

- Catalysts
    - Helps remove any residual oxygen
    - Palladium-coated alumina pellets

- Desiccant
    - Absorbs excess water produced when catalyst removes oxygen to form water

- Indicator
    - Colorless when no oxygen present
    - Blue or pink when oxygen present, depending on dye

- Incubation time (5-7 days)
    - Anaerobic cultures are routinely held 5 to 7 days to allow growth of particularly slow-growing anaerobes and at least 10 days whenever *Actinomyces* is suspected
    - Incubated at 35°C to 37°C

Glove boxes

- Available as sealed glove boxes
    - Glove boxes are fitted with airtight rubber gloves. Easier to use and less susceptible to oxygen contamination than gloveless chambers

- Gloveless chambers
    - Uses airtight rubber sleeves and ccommodates many different hand sizes

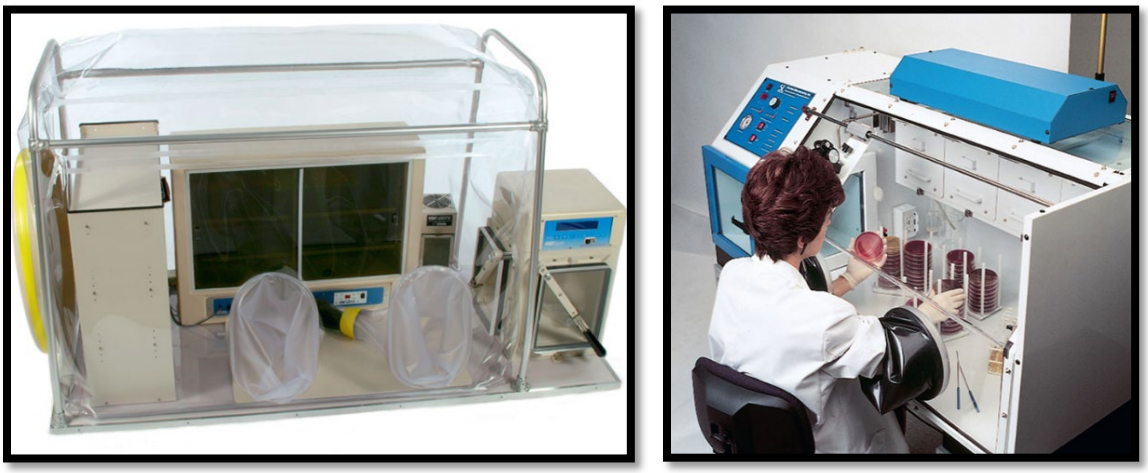


Figure 7.7 – Glove Boxes

Anaerobe Jars and Anaerobic Bags

- Anaerobe jars
    - Good for small laboratories
    - Low volume of anaerobic cultures
    - Newer models accommodate a larger number of plates and microtiter susceptibility trays and anaerobic identification strips or trays

- Anaerobic bags
    - One or two inoculated plates are placed into a bag, an oxygen removal system is activated, and the bag is sealed and incubated



Figure 7.8 – Anaerobe Jars and Anaerobic Bags

Options Available for Identifying Anaerobic Bacteria

- Presumptive identification - based on colony morphology, Gram stain observations, and results of simple tests (e.g., disks, catalase, spot indole)
    - Limited capability; many clinically encountered anaerobes cannot be identified

- Definitive identification commercially available, preexisting, enzyme-based identification systems (e.g., ANIDENT, microscan, rapid ANA-II, BBL Crystal, Vitek ANI card)
    - Most commonly encountered, clinically significant anaerobes can be identified

Indications of the Presence of Anaerobes in Cultures

- Foul odor when opening an anaerobic jar or bag

- Characteristic colonial morphology on anaerobic agar plates but not on aerobic culture plates

- Unique morphology on Gram stain of clinical specimens or isolates

- Good growth on Bacteroides bile-esculin agar (BBE)

- Double zone of hemolysis on blood agar

- Brick-red fluorescence on kanamycin vancomycin laked blood (KVLB) agar

Processing Colonies Suspected of Being Anaerobes

- Describe colony morphology and growth on selective media

- Describe the gram-stain reaction and cell morphology

- Set up an aerotolerance test

- Inoculate pure culture/subculture plate and add appropriate disks

- It is recommended to magnify the plate to observe the fine details of each colony morphotype and to pick colonies for isolation



Figure 7.9 – Magnifying Lamp

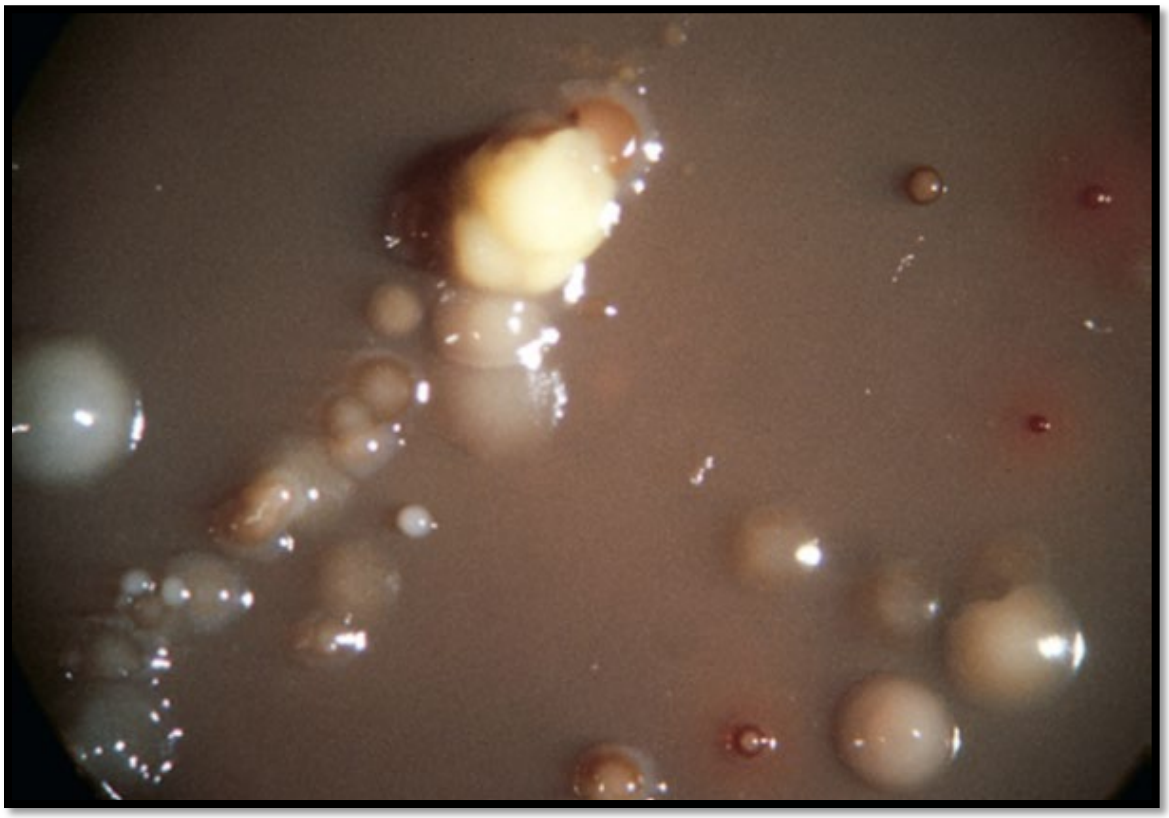


Figure 7.10 – Depicts the appearance of colonies as seen through a dissection microscope.

Gram Stain

- The gram stain reaction and morphologic appearance of the organism aid in the presumptive identification of isolates

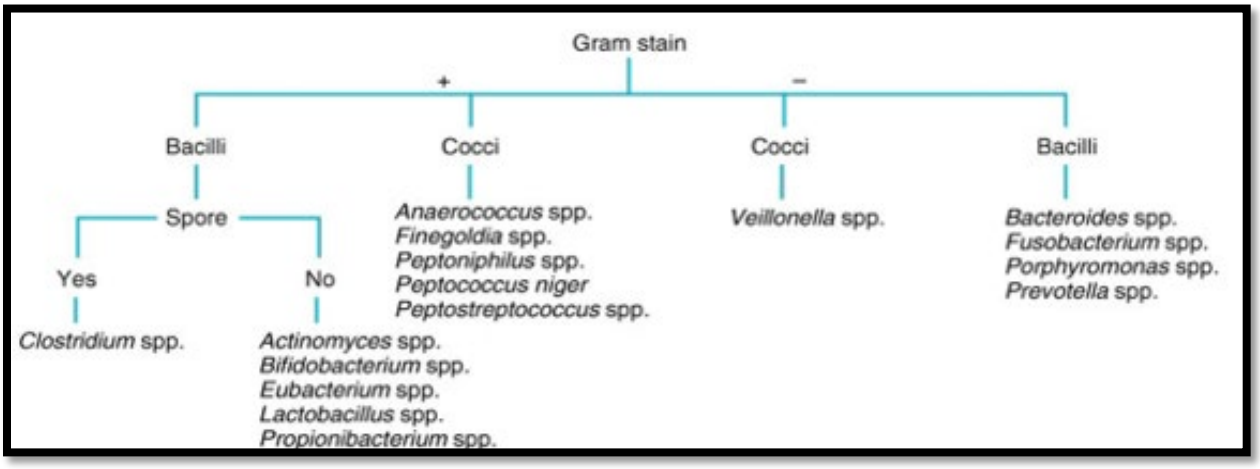


Figure 7.11 – Schematic Diagram for Initial Identification of Anaerobic Isolates Based on Gram-Stain Morphology

Interpretation of Aerotolerance Test Results

- Aerotolerance test
  - Determines whether a microorganism isolated under anaerobic conditions is a strict anaerobe or a facultative anaerobe
  - Inoculated with a short streak onto a blood and chocolate agar plate for incubation in a CO_2 incubator and an anaerobic blood agar(CDC) incubated anaerobically
  - Some aerotolerant anaerobes (e.g., certain *Clostridium*, *Actinomyces*, *Propionibacterium*, and *Lactobacillus* spp.) Can grow on the CO_2-incubated chocolate agar, but they usually grow much better on the anaerobically incubated plate

Table 7.4 – Interpretation of Aerotolerance Test Results

	Aerobe	Capnophilic Aerobe	Facultative Anaerobe	Obligate Anaerobe
Blood agar plate incubated aerobically in a non-CO_2 incubator	+	–	+	–
Chocolate agar plate incubated aerobically in a CO_2 incubator	+	+	+	–
Blood agar plate incubated anaerobically	–	–*	+	+

Presumptive Identification

- Rapid tests
  - Fluorescence
  - Catalase test
  - Spot indole test
  - Urease tests
  - Motility test
  - Special Potency Antimicrobial Disks
  - Lecithinase, lipase, and proteolytic reactions

Rapid Tests

- Fluorescence
  - Ultraviolet (UV) light
    - *Porphyromonas* and *Prevotella* spp.
      - Brick-red fluorescence
    - *Fusobacterium nucleatum* and *Clostridium difficile*
      - Chartreuse fluorescence
    - *Veillonella* spp.
      - Red fluorescence
        - Dependent on culture medium
        - Fades rapidly

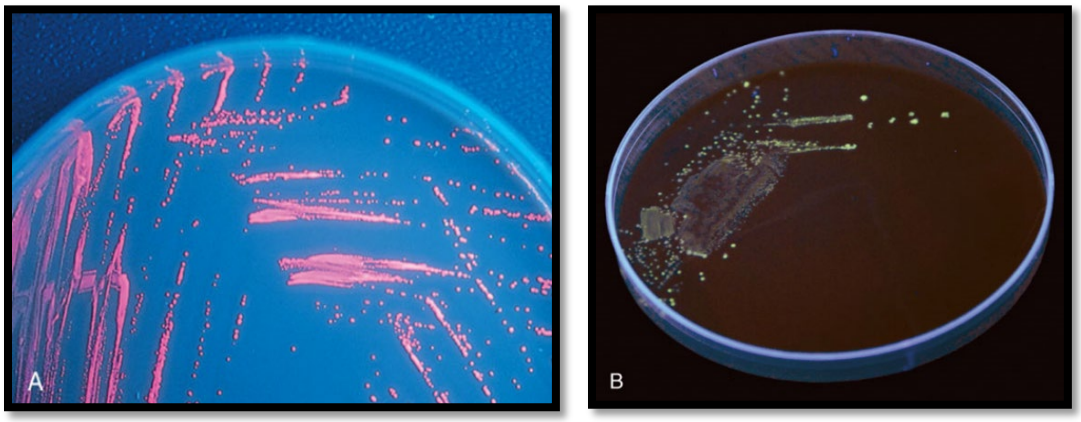


Figure 7.12 – Fluorescent Bacteria under UV light

Table 7.5 – Fluorescence Under Long-Wave UV Light

Organism	Color of Fluorescence
Prevotella (pigmented)	Brick red
P. bivia, P. disiens	Light orange to pink (coral)
Porphyromonas	Brick red (some no fluorescence)
Fusobacterium	Chartreuse
Veillonella	Red but fades rapidly
C. innocuum, C. difficile	Chartreuse
Eggerthella lenta	Red or no fluorescence

- Catalase test
    - 15% hydrogen peroxide added to organism on slide
        - Bubbling is positive
            - Separates *Clostridium* (negative) from *Bacillus* (positive)

- Spot indole test
    - Dilated cardiomyopathy with ataxia (DCMA) reagent
        - Filter paper saturated with para-dimethylaminocinnamaldehyde
        - Turns a blue or green color (positive)
        - Pink or orange color (negative)

- Rapid urease test 2-3 hours
    - Urease Test Tablet to 1 ml of distilled water then heavily inoculated into the water and incubated at 37°C
        - Positive reaction can occur 15 minutes to 6 hours indicated by a color change from yellow to red

- Rapid motility test as soon as 18 hours
    - Wet mount using either (4 to 6 hours old) broth cultures or 24-48 hour old colonies
    - A small portion of a colony should be touched to a drop of saline on a microscope slide. A coverslip is carefully placed over the drop, and the slide is examined with a light microscope
        - Positive motility test is indicated when organisms exhibit purposeful movement

- Special Potency Disks
    - To determine the true gram stain reaction of the isolate, special potency disks can be used
    - Vancomycin disk
        - Susceptibility indicates gram-positive bacillus or *Porphyromonas*
    - Kanamycin disk
        - Susceptibility indicates *Bacteroides* other than *fragilis* group
        - Also *Bilophila, Fusobacterium, Veillonella*
    - Colistin disk
        - Susceptibility indicates same as kanamycin disk plus *Prevotella*
    - Nitrate disk
        - Ability to reduce nitrate
    - Sodium polyanethol sulfonate (SPS) disk
        - *Peptostreptococcus anaerobius* is susceptible
        - *Peptoniphilus asaccharolyticus* is resistant
    - Bile disk
        - Growth in 20% bile
        - *B. fragilis* group

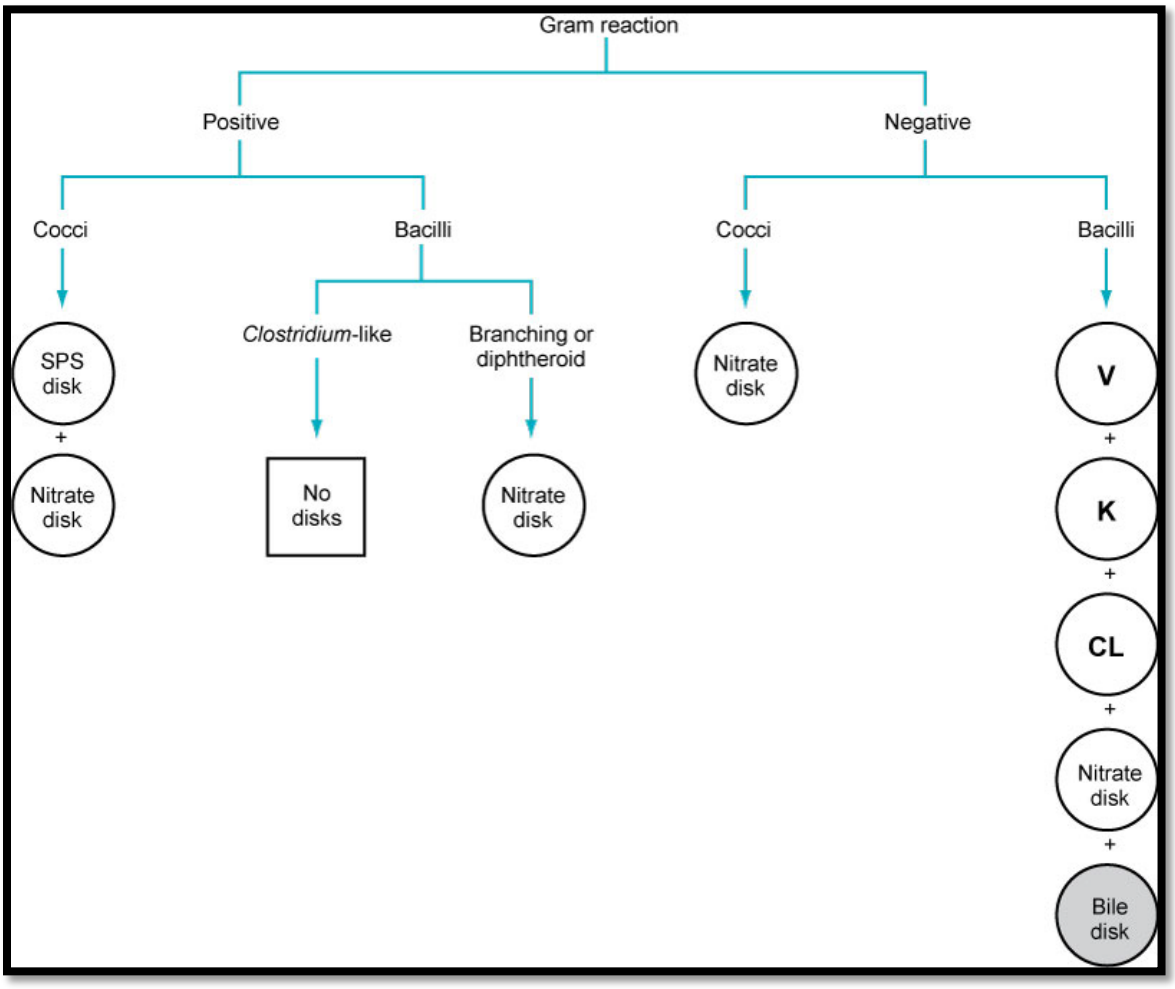


Figure 7.13 – Interpretation of Special-Potency Antimicrobial Disk Results

Table 7.6 – Interpretation of Special-Potency Antimicrobial Disk Results

Vancomycin	Kanamycin	Colistin	Interpretation
S	V	R	• Probably a pink-staining, gram-positive bacillus such as *Clostridium ramosum* or *C. clostridioforme*; however, if the kanamycin result is resistant, it could be a *Porphyromonas* sp.
S	R	R	• *Porphyromonas* sp.
R	R	R	• Probably a member of the *Bacteroides fragilis* group, but could be a *Prevotella* sp.
R	R	V	• *Prevotella* sp.
R	R	S	• Probably a *Prevotella* sp.
R	S	S	• Bacteroides ureolyticus, Bilophila wadsworthia, or a Fusobacterium sp.; Veillonella



Figure 7.14 – Interpretation of Special-Potency Antimicrobial Disk Results

- Lecithinase and Lipase Reactions
    - Egg yolk agar (EYA)
        - These reactions are of value in identifying many species of *Clostridia*
        - Lecithinase cleaves lecithin found in egg yolk
            - Releases insoluble fat
                - Forms opaque zone around colony
                - *Clostridia perfringens*
        - Lipase
            - Hydrolyzes triglycerides and diglycerides to fatty acids and glycerol
                - Iridescent, multicolored sheen
                - *Fusobacterium necrophorum*

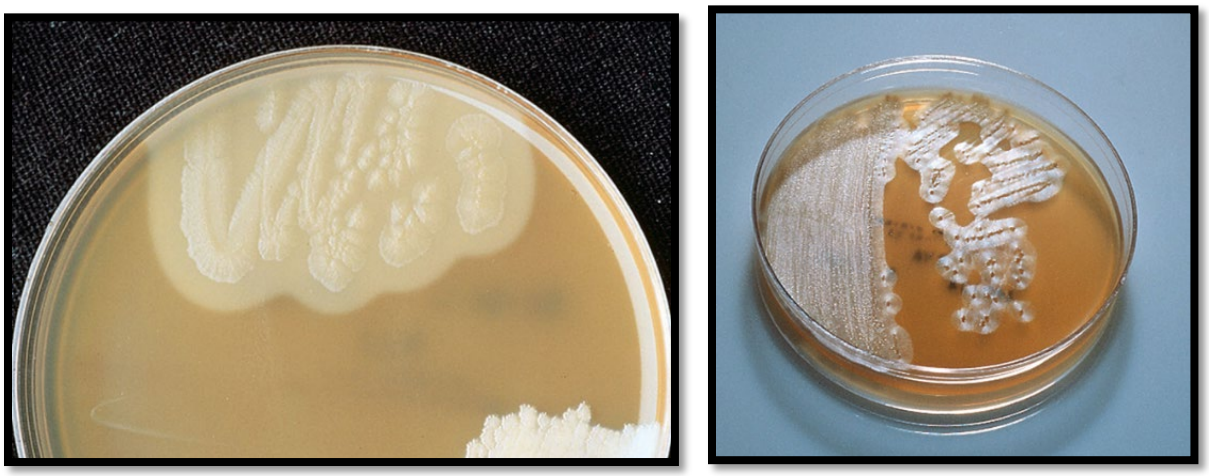


Figure 7.15 – Positive Lecithinase and Lipase Reactions

Gram-positive, Spore-Forming Anaerobic Bacilli

- All spore-forming anaerobic bacilli are classified in the genus *Clostridium* and are collectively termed *clostridia*
    - Grouped according to the location of the endospore within the cell

***Clostridium* Species**

- *Clostridium difficile* - Thin rods, rare spores
- *Clostridium perfringens* - Boxcar, large, square rods
- *Clostridium septicum* - Thin rods, subterminal spores
- *Clostridium sordellii* - Thin rods, subterminal spores
- *Clostridium tetani* - Swollen terminal spores

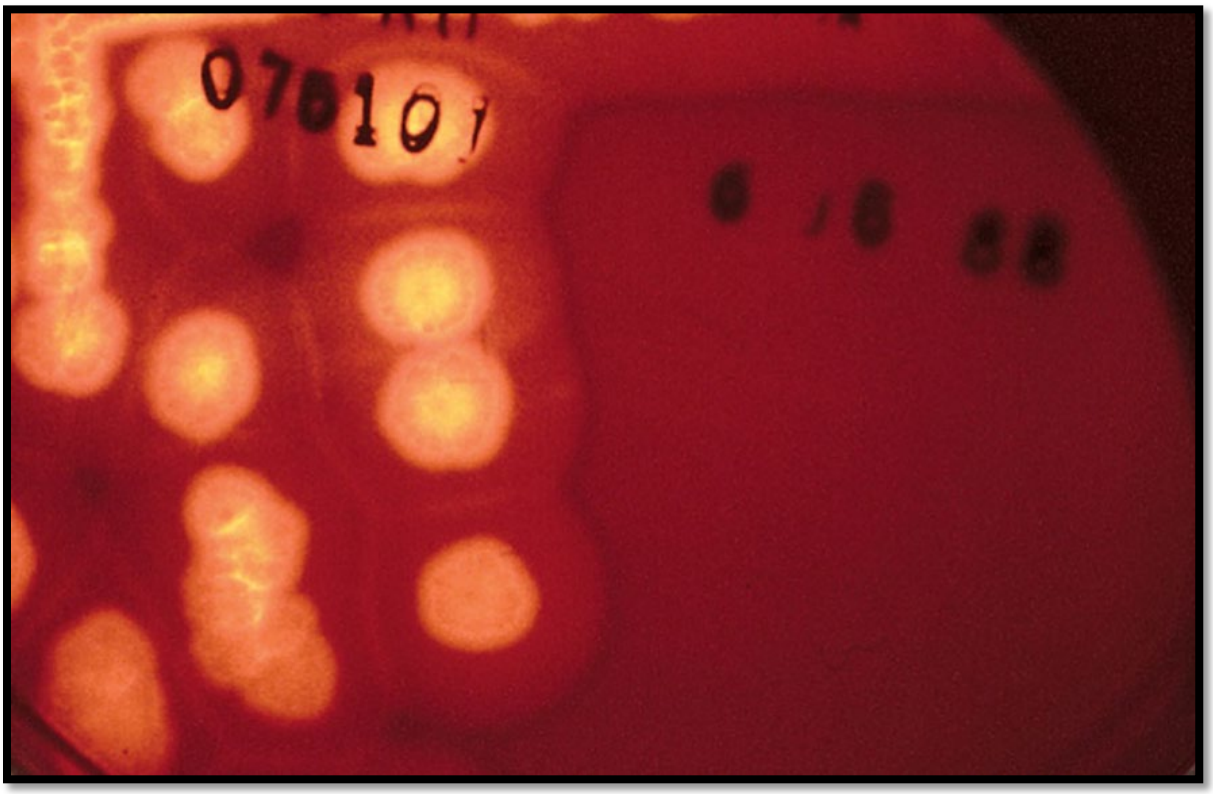


Figure 7.16 – C. perfringens double zone of hemolysis

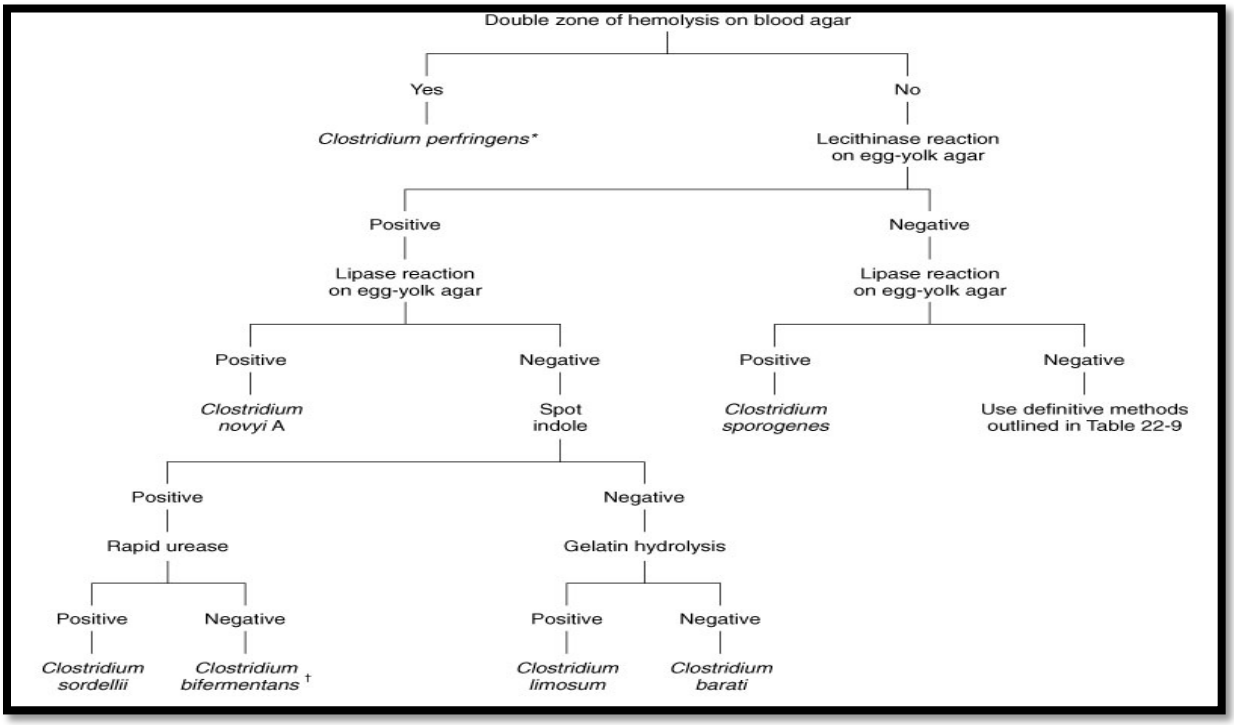


Figure 7.17 – Identification Algorithm for Some Clinically Encountered Clostridium Species

Identification of Non–Spore-Forming Gram-positive Bacilli

- *Actinomyces*
  - Straight to slightly curved short rods to long filaments
  - Short rods with clubbed ends in diphtheroid arrangements
  - Short chains, or small clusters
  - Longer rods and filaments may be straight or wavy and branched

- *Bifidobacterium*
  - Variable in shape, ranging from coccobacilli to long branching rods. The ends of the cells may be pointed, bent, club-shaped, spatulated, or bifurcated

- *Lactobacillus*
  - Highly pleomorphic may appear coccoid or spiral-shaped

- *Propionibacterium*
  - Pleomorphic rods with a diphtheroid appearance

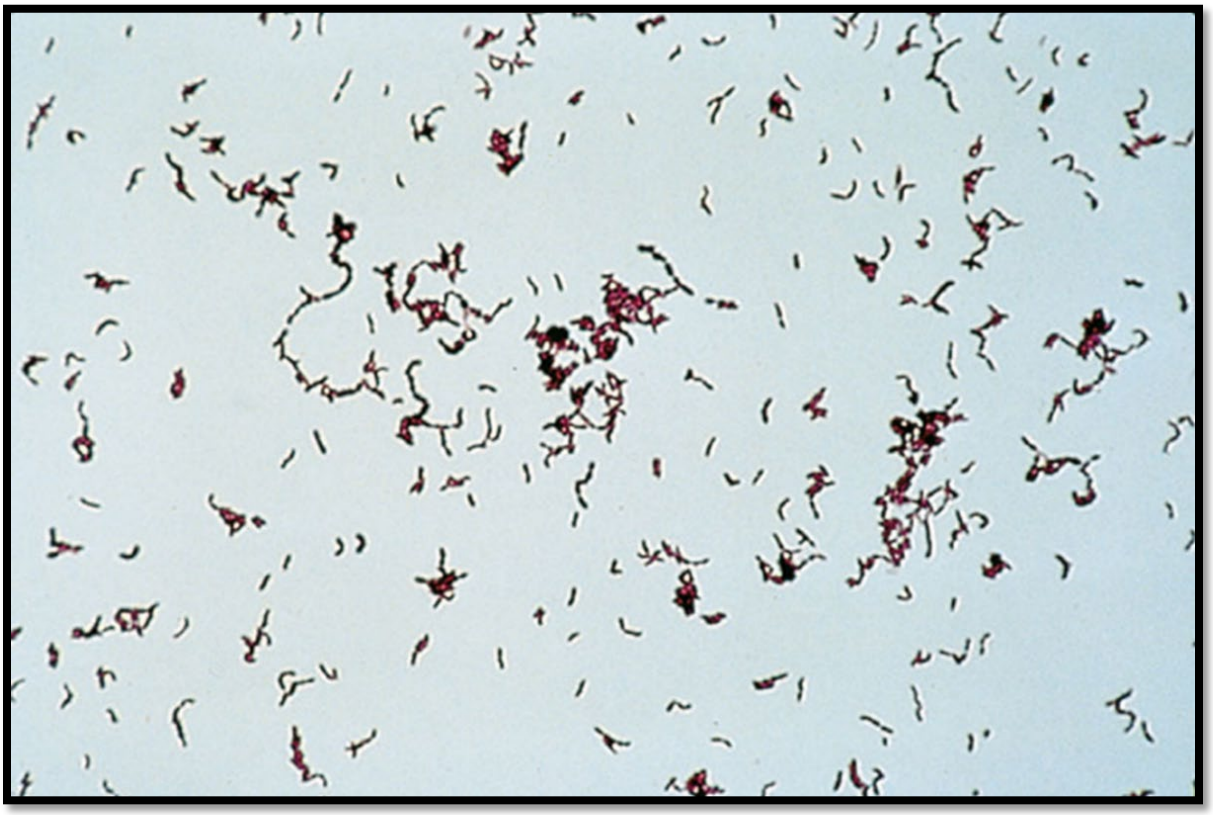


Figure 7.17 – Propionibacterium acnes

Table 7.7 – Characteristics of Gram-positive Non–Spore-Forming Anaerobic Bacilli

	48-h Colony (<1 mm)	Red Pigment Colony	β-Hemolytic	Rough Colony	Branched Bacilli	Catalase	Indole	Comments
Actinomyces israelii	+	–	–	+	+	–	–	**Molar tooth colony**
A. Meyeri	–+	–	–	–	+	–	–	
A. naeslundii	–+	+	–	–+	+	–	–	
A. odontolyticus	–	–	–	–	+	–	–	
A. viscosus	–	–	–	–+	+	+	–	

	48-h Colony (<1 mm)	Red Pigment Colony	β-Hemolytic	Rough Colony	Branched Bacilli	Catalase	Indole	Comments
Bifidobacterium spp.	–	–	–	–	–	–	–	**Rods with forked ends**
Eggerthella lenta	–	–	–	–	–	V	–	**Nitrate-positive**
Eubacterium spp.	–	–	–	–	–	–	–	
Propionibacterium acnes	–	–	–	–	–	+	+	
P. propionicum	+	–	–	+	+	–	–	

Identification of Anaerobic Gram-negative Bacilli

- *Bacteroides* spp.
- *Bilophilia* spp.
- *Prevotella* spp.
- *Porphyromonas* spp.
- *Fusobacterium*

***Bacteroides* spp.**

- *B. fragilis* group
    - Bile resistant
        - KVLB agar plate
        - BBE agar plate
    - *B. fragilis*
        - Most common species of anaerobic bacteria isolated from infectious processes of soft tissue and anaerobic bacteremia
        - Responsible for more than 60% of the infections caused by the *B. fragilis* group
    - *B. caccae*
    - *B. thetaiotaomicron*
        - Second most frequently encountered of group
        - Highest degree of antimicrobial resistance of the group
    - *B. uniformis*
        - Nonpigmented
        - Esculin Hydrolysis negative

- *Bacteroides ureolyticus*
    - Bile sensitive
    - Nonpigmented
    - Isolated from blood following tooth extraction
    - Urease positive
    - Tiny rods or coccobacilli

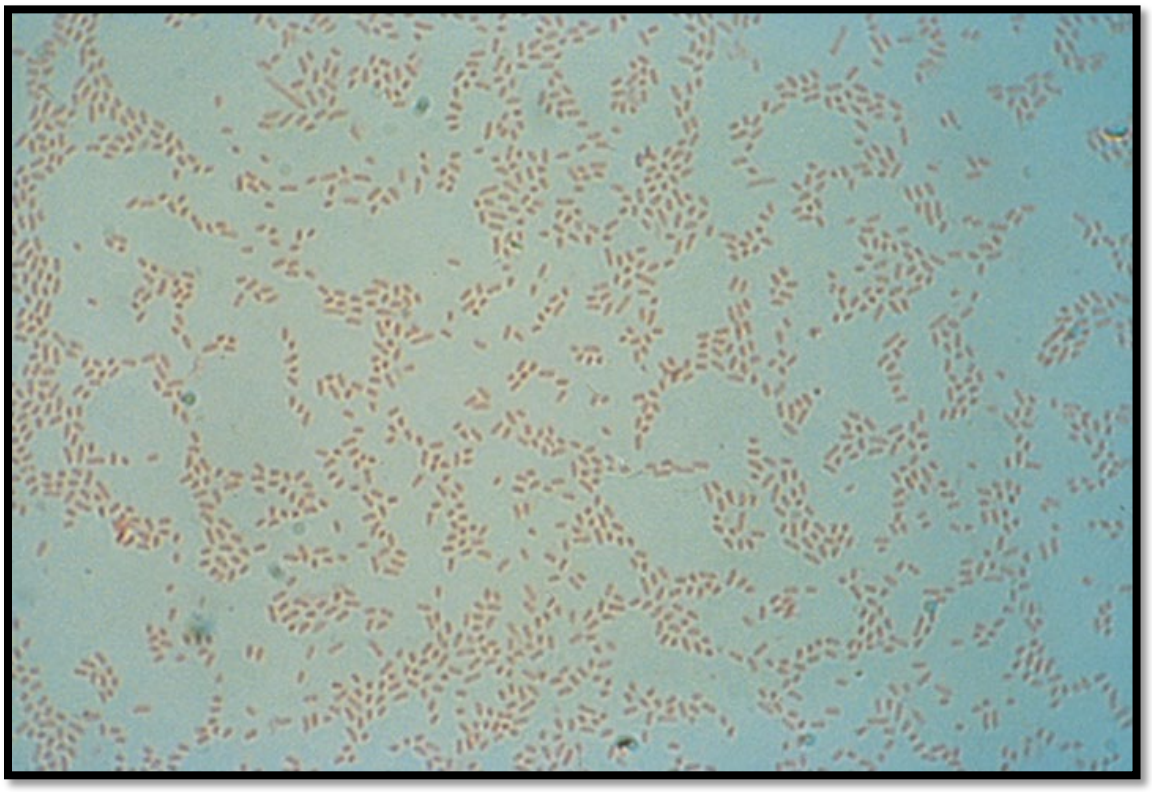


Figure 7.18 – B. fragilis

***Bilophilia* spp.**

- Bilophila wadsworthia
    - Bile resistant
    - Pigmented
    - BBE agar characteristic fish-eye appearance

***Prevotella* spp.**

- *Prevotella* spp.
    - Bile sensitive
    - Pigmented
        - Produce protoporphyrin, causing a brown-black color shift of macroscopic colonies over 2 to 3 weeks
    - Brick-red fluorescence under long-wave UV light
    - Tiny coccobacilli or bacilli
    - Vagina and oral cavities
    - *P. Denticola* esculin positive
    - *P. Intermedia* indole positive, lipase positive
    - Some bile-sensitive pigmented species can be categorized as pitting or nonpitting colonies

***Porphyromonas* spp.**

- *Porphyromonas* spp.
  - Bile sensitive
  - Pigmented
  - Brick-red fluorescence under long-wave UV light
  - Tiny coccobacilli
  - No growth on media containing vancomycin

***Fusobacterium* spp.**

- Generally considered gram-negative long, thin, tapered rods (fusiform)
  - May appear pleomorphic

- *F. nucleatum*
  - Bile sensitive
  - Classic fusiform shape
  - Indole positive
  - Butyric acid from peptone yeast glucose (PYG) media

- *F. mortiferum*
  - Bile resistant
  - Indole negative
  - Esculin positive

- *F. russi*
  - Bile sensitive
  - Indole negative



Figure 7.19– Fusobacterium spp.



Figure 7.20 – F. nucleatum

Table 7.8 – Characteristics of Anaerobic Gram-negative Bacilli

	Vancomycin	Kanamycin	Colistin	Bile Resistant	Growth on KVLB	Catalase	Formate- Fumarate	Comments
B. fragilis	R	R	R	+	+	+		
B. ureolyticus	R	S	S	–	–	V	+	Pitted and nonpitted
Fusobacterium nucleatum	R	S	S	–	–	+		Chartreuse Fluorescence Slender, pointed cells
F. mortiferum	R	S	S	+		–		Chartreuse Fluorescence Bizarre, round bodies

Identification of Anaerobic Cocci

- *Peptococcus niger*
- *Peptostreptococcus anaerobius*
- *Peptoniphilus asaccharolyticus*
- *Veillonella*

Anaerobic Cocci

- Many clinical specimens may contain these organisms
    - Vagina, abscesses are most likely sites
        - Cocci in a variety of arrangements
            - Tetrads, chains, clusters
- *Peptococcus niger*
    - *Black pigmentation*
- *Peptostreptococcus anaerobius*
    - SPS susceptible
    - Not black pigmented
- *Peptoniphilus asaccharolyticus*
    - SPS resistant
    - Indole positive
    - Does not reduce nitrate
    - Not black pigmented
- *Veillonella*
    - Gram-negative cocci
    - Positive catalase
    - Red fluorescence

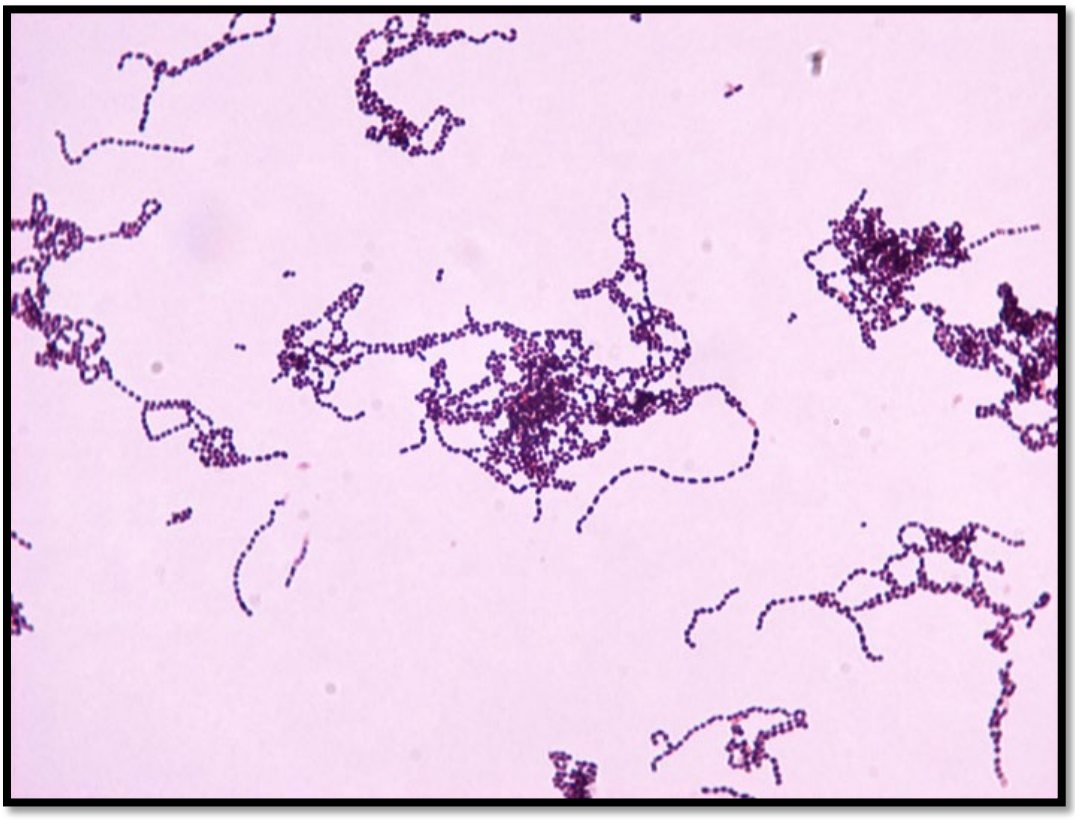


Figure 7.21– Gram Stain of Anaerobic Gram-positive Cocci

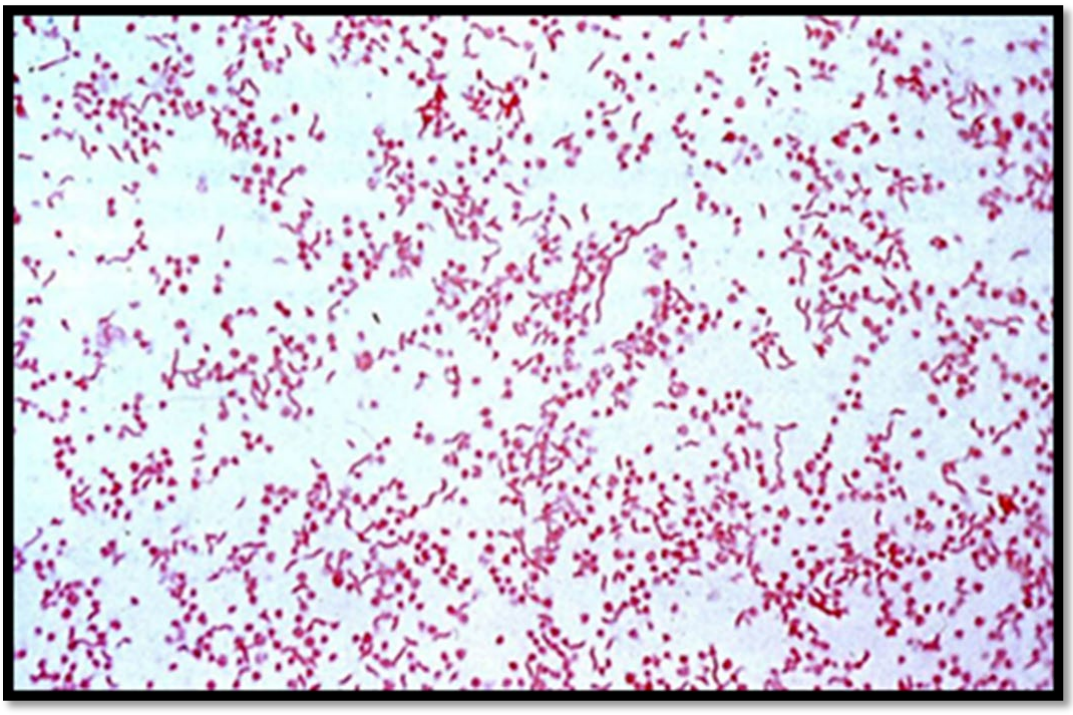


Figure 7.22 – Gram Stain of Anaerobic Gram-negative Cocci Veillonella

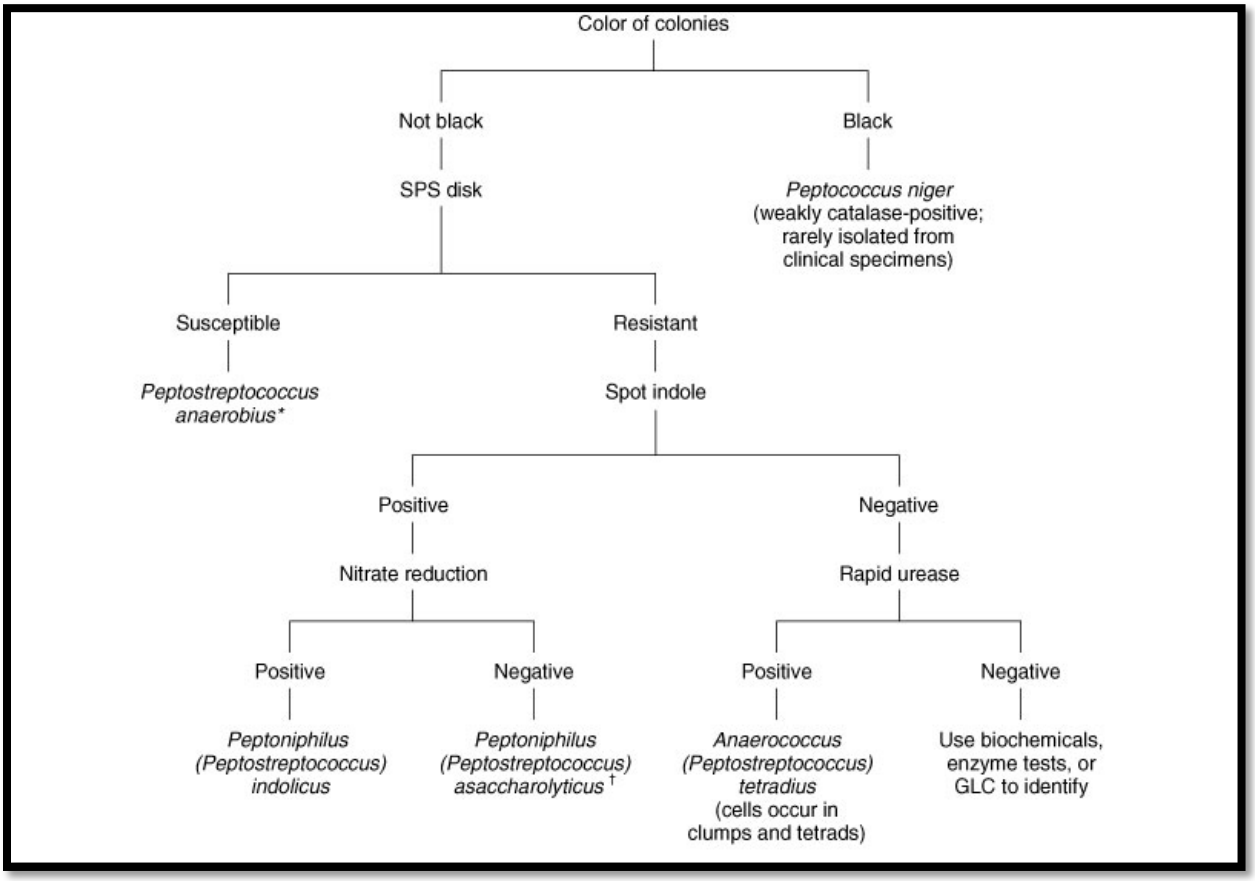


Figure 7.23 – Identification of Anaerobic Gram-positive Cocci

Table 7.9 – Characteristics of Some Clinically Encountered Anaerobic Cocci

	Gram-Stain Reaction	Catalase	Indole	Urease	Nitrate Reduction	Red Fluorescence	Major Products in PYG by GLC	Comments
Peptostreptococcus anaerobius	+	–	–	–	–*	–	A, IB, IV, IC	Inhibited by SPS disk
Parvimonas micra	+	–	–	–	–	–	A	
Peptoniphilus asaccharolyticus	+	–*	+	–	–	–	A, B	
Peptoniphilus indolicus	+	–	+	–	+	–	A, P, B	
Finegoldia magna	+	–*	–	–	–	–	A	
Anaerococcus prevotii	+	–*	–	–	–	–	A, P, B	
Anaerococcus tetradius	+	–	–	+	–	–	A, B, L	
Veillonella parvula	–	+	–	–	–*	+*	A, P	

Modified from Engelkirk PG et al: *Principles and practice of clinical anaerobic bacteriology*, Belmont, CA, 1992, Star.
A, Acetic acid; *B*, butyric acid; *GLC*, gas liquid chromatography; *IB*, isobutyric acid; *IC*, isocaproic acid; *IV*, isovaleric acid; *L*, lactic acid; *PYG*, peptone–yeast extract–glucose; *P*, propionic acid; *SPS*, sodium polyanethol sulfonate; +*, most strains positive; –*, most strains negative.

Table 7.10 – Presumptive Identification of Gram-positive Anaerobes

Identification	Colony Morphology on Blood Agar	Cellular Morphology
• *Clostridium difficile*	• Large, flat colonies; barnyard odor, chartreuse fluorescence	• Thin rods, rare spores
• *C. perfringens*	• Large, irregular-shaped, double zone of β-hemolysis	• Boxcar, large, square rods
• *C. septicum*	• Smoothly swarming	• Thin rods, subterminal spores Negative
• *C. sordellii*	• Very large, lobate, irregular, flat	• Thin rods, subterminal spores

Identification	Colony Morphology on Blood Agar	Cellular Morphology
• *C. tetani*	• Smoothly swarming but slow growing	• Swollen terminal spores
• Peptostreptococci	• Small, peaked, circular	• Cocci, pairs and chains
• *Propionibacterium acnes*	• Small, opaque, enamel white, circular (catalase-positive)	• Coryneform rods

Table 7.11 – Presumptive Identification of Gram-negative Anaerobes

Identification	Colony Morphology on Blood or KVLB Agar	Colony Morphology on BBE	Cellular Morphology
• *B. fragilis* group	• Large (>1 mm)	• Large, convex, black-gray	• Regular
• *B. ureolyticus*	• Translucent, pitting (some)	• No growth	• Tiny rods or coccobacilli
• *Bilophila wadsworthia*	• Tiny, translucent	• Translucent, with black center at 72h	• Regular to filamentous Negative
• *Fusobacterium nucleatum*	• Ground-glass or breadcrumb	• No growth	• Fusiform, thin with pointed ends

Identification	Colony Morphology on Blood or KVLB Agar	Colony Morphology on BBE	Cellular Morphology
• *Porphyromonas*	• Small, translucent or opaque, brick red fluorescence on blood agar, no growth on KVLB	• No growth	• Tiny coccobacilli
• *Prevotella intermedia*	• Small, translucent or opaque, brick red fluorescence on blood and KVLB agar	• No growth	• Tiny coccobacilli
• *Veillonella*	• Small, translucent or opaque, brick red fluorescence on blood agar, no growth on KVLB	• No growth	• Tiny diplococci

Definitive Identifications

- Biochemical and enzyme-based minisystems
    - Based on the presence of preformed (preexisting) bacterial enzymes
        - Systems do not depend on enzyme induction or bacterial growth
    - Systems are primarily of value for identifying only the most commonly isolated anaerobes
    - Vital to have an accurate gram stain to arrive at an accurate identification

- Conventional Tubed Biochemical
    - Identification Systems PRAS and non-PRAS tubed biochemical test media
        - PRAS media are prepared by boiling, autoclaving, and and replacing any air with an oxygen-free gas mixture
        - Biochemical reactions are interpreted by measuring the pH with a probe or by color changes based on a bromthymol blue indicator
    - Gas-Liquid Chromatographic
        - Anaerobe to be tested is grown in a tube of peptone–yeast extract–glucose (PYG) medium

- Cellular fatty acid analysis by High-Resolution Gas-Liquid Chromatography (GLC)
    - Are coded for on bacterial chromosomes and not affected by simple mutations or plasmid loss

- 16S Ribosomal RNA Gene sequencing
    - Has highly conserved and highly variable regions that can be used for microbial identification

Topic: Antimicrobial Susceptibility Tests

Topic Objective: Discuss antimicrobial susceptibility testing of anaerobes

Antimicrobial Susceptibility Testing

- Clinical and Laboratory Standards Institute (CLSI)
    - CLSI document provides guidelines for susceptibility testing of anaerobes
        - *Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria* (Approved Standard, M11-A8)

- Performed on
    - Brain abscess
    - Endocarditis
    - Infection of a prosthetic device or vascular graft
    - Joint infections
    - Osteomyelitis
    - Bacteremia

- Highly virulent and/or commonly resistant to antimicrobial agents
    - *Bacteroides fragilis* group
    - *Prevotella*
    - *Clostridium*
    - *Fusobacterium*
    - *Bilophila wadsworthia*
    - *Sutterella wadsworthensis*

CLSI Recommendations for the Antimicrobial Susceptibility testing

- β-Lactamase tetsing is performed on all anaerobes

- Gram-negative Anaerobes (Including *B. fragilis* group)
    - Primary choices
        - β-Lactam combination agent (e.g., ampicillin-sulbactam, piperacillin-tazobactam, ticarcillin-clavulanic acid)
        - Clindamycin
            - Carbapenem (e.g., ertapenem, imipenem, meropenem)
            - Metronidazole
    - Supplemental choices
        - Ceftizoxime, ceftriaxone
            - Cefotetan, cefoxitin
            - Moxifloxacin

- Gram-positive Anaerobes
    - Primary choices

        - β-Lactam combination agent (e.g., ampicillin-sulbactam, piperacillin-tazobactam, ticarcillin-clavulanic acid)
        - Clindamycin
            - Carbapenem (e.g., ertapenem, imipenem, meropenem)
            - Metronidazole, ampicillin or penicillin
    - Supplemental choices
        - Ceftizoxime, ceftriaxone
            - Tetracycline
            - Moxifloxacin

Problems in Susceptibility Testing of Anaerobic Isolates

- Issues
    - Of greatest concern are the accuracy of susceptibility methods and there correlation with in vivo or clinical situation
    - Cost and procedure complexity
    - Difficulty in reading endpoints with certain susceptibility methods
    - Lack of reproducibility
    - Failure of some anaerobes to grow on or in particular media
    - Lack of comparability between methods

Topic: Treatment

Topic Objective: Describe treatment of anaerobe-associated diseases

Treatment of Anaerobe-Associated Diseases

- Surgical Therapy
  - Surgical drainage of the abscess
  - Débridement
  - Eliminating obstructions
  - Decompressing tissues
  - Releasing trapped gas
  - Improving circulation in, and oxygenation of, tissues

- Hyperbaric oxygen chamber
  - Used to force oxygen into necrotic tissue

- Antimicrobial Therapy
  - The primary role to limit the local and systemic spread of the organisms
  - Selection of the correct antimicrobial agent depends on these factors
    - Types of organisms involved
    - Known resistance factors
    - Site of the infection
    - Toxicity of the antimicrobial agent

- Antitoxins to reduce virulence factors
  - In cases of tetanus and botulism, antitoxins are used to neutralize the effect of neurotoxins produced by *C. tetani* and *C. botulinum*

Checks on learning

1. An anaerobe is an organism that:

2. What substances do organisms that use oxygen have to protect themselves from superoxide anions?

3. What are two important factors that must be taken into consideration when transporting specimens for anaerobic culture?